

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(END SEMESTER EXAMINATION)

CLASS: IMSC
BRANCH: MATHS

SEMESTER : I
SESSION : MO/2023

SUBJECT: MA101 CALCULUS - I

TIME: 3 Hours

FULL MARKS: 50

INSTRUCTIONS:

1. The question paper contains 5 questions each of 10 marks and total 50 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.

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|--|---------|----|
| Q.1(a) If $\cos^{-1}\left(\frac{y}{b}\right) = \log\left(\frac{x}{n}\right)^n$, prove that $x^2y_{n+2} + (2n+1)xy_{n+1} + 2n^2y_n = 0$ | [5] CO1 | 2 |
| Q.1(b) Discuss the applicability of Rolle's theorem on the function given by
$f(x) = \begin{cases} x^2 + 1, & \text{if } 0 \leq x \leq 1 \\ 3 - x, & \text{if } 1 \leq x \leq 2. \end{cases}$ | [5] CO1 | 1 |
| Q.2(a) Find the asymptotes of $x^3 + y^3 = 3axy$ | [5] CO2 | 2 |
| Q.2(b) Show that for the curve $y = \frac{ax}{a+x}$, the radius of curvature ρ at (x, y) is related as
$\left(\frac{2\rho}{a}\right)^{2/3} = \frac{x^2}{y^2} + \frac{y^2}{x^2}$ | [5] CO2 | 2 |
| Q.3(a) Expand $x^2y + 3y - 2$ in powers of $x - 1$ and $y + 2$ using Taylor's theorem | [5] CO3 | 1 |
| Q.3(b) Discuss the maxima and minima of $f(x, y) = x^3y^2(1 - x - y)$ | [5] CO3 | 2 |
| Q.4(a) Prove that $\int_{\frac{\pi}{2}-\alpha}^{\frac{\pi}{2}} \sin \theta \cos^{-1}(\cos \alpha \operatorname{cosec} \theta) d\theta = \frac{\pi}{2}(1 - \cos \alpha)$. | [5] CO4 | 3 |
| Q.4(b) Show that $\int_0^1 \frac{x^2 dx}{(1-x^4)^{1/2}} \cdot \int_0^1 \frac{dx}{(1-x^4)^{1/2}} = \frac{\pi}{4}$. | [5] CO4 | 2 |
| Q.5(a) Find the area enclosed between parabolas $y^2 = 4a(x + a)$ and $y^2 = -4a(x - a)$ | [5] CO5 | 2 |
| Q.5(b) Find the volume of the solid generated by revolving the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about the major axis | [5] CO5 | 3 |

:::11/12/2023 M:::